AURORA Proteus-to-Real-EV Mapping

What each simulated element represents in an actual electric vehicle

System Mapping

Proteus Element	Current Function	Real-EV Equivalent
BMS temperature potentiometer	Creates the battery-temperature input used for BMS state decisions.	Pack/module temperature sensor or thermistor input.
BMS STM32F401RE	Runs BMS state and torque-limit logic.	Battery Management System ECU / battery supervisory controller.
Green / red BMS LEDs	Show OK/fault-related BMS state.	Diagnostic/status indicator or service-tool status.
ACC potentiometer	Represents accelerator request to VCU.	Accelerator Pedal Position Sensor (APPS).
BRK potentiometer/input	Represents brake request.	Brake pedal position / brake request sensor.
SOC potentiometer on VCU	Represents state-of-charge input used by VCU logic.	SOC value normally estimated by the BMS and sent to the VCU.
Ignition/start switch	Enables vehicle operation.	Ignition/start request or vehicle enable signal.
Charge switch	Indicates charging connection/state.	Charge-port / EVSE connection or charging-enable signal.
VCU STM32F401RE	Runs supervisory vehicle-state and torque arbitration logic.	Vehicle Control Unit.
UART links: BMS-VCU / VCU-Motor	Transfers battery state, motor command and health.	In-vehicle network communication, commonly CAN in production.
Motor Controller STM32F401RE	Receives motor command and generates PWM.	Motor Control Unit / inverter control ECU.
TIM3_CH1 PWM	Controls MOSFET duty cycle.	Gate-control command used by an inverter power stage.
Single MOSFET + DC motor	Simplified traction actuation demo.	Traction inverter plus PMSM/BLDC traction motor and drivetrain.
Motor fault switch	Injects a simulated motor-side hardware fault.	Inverter/motor diagnostic fault feedback.
Virtual Terminals	Display internal state and DTCs.	Dashboard, diagnostics interface or service tool.

What the Proteus Prototype Demonstrates Well

- Separation of responsibilities between BMS, VCU and Motor Controller.
- Battery protection affecting available torque.
- Driver input processing and vehicle-state decisions.
- Final torque-command generation and hardware PWM actuation.
- Fault feedback from the Motor Controller to the VCU.
- End-to-end ECU communication and diagnostics.

What Is Simplified Compared with a Real EV

- The motor stage is a single-MOSFET DC-motor demonstration rather than a three-phase traction inverter.
- Regenerative braking is logical rather than physically bidirectional.
- The BMS does not presently model cell-by-cell voltage sensing, balancing or current measurement.
- UART is used instead of a production automotive communications network.

- The detailed PMSM/FOC/EKF work remains in MATLAB/Simulink rather than being fully represented in the Proteus motor power stage.